Porter Logistics Case Study

About Porter

Porter is India's Largest Marketplace for Intra-City Logistics. Leader in the country's $40 billion intra-city logistics market, Porter strives to improve the lives of 1,50,000+ driver-partners by providing them with consistent earning & independence. Currently, the company has serviced 5+ million customers

Business Problem

Porter works with a wide range of restaurants for delivering their items directly to the people. Porter has a number of delivery partners available for delivering the food, from various restaurants and wants to get an estimated delivery time that it can provide the customers on the basis of what they are ordering, from where and also the delivery partners. Porter wants to estimate the delivery time for a given order and improve the customer experience by providing accurate delivery time estimates. The company has provided a dataset containing information about past orders, including the time taken for each delivery. The goal is to build a regression model that can predict the delivery time based on various features such as order details, restaurant information, and delivery partner availability. This dataset has the required data to train a regression model that will do the delivery time estimation, based on all those features

Metrics

We will use Mean Squared Error (MSE) as the evaluation metric for our regression model. MSE is a common metric used to measure the average squared difference between the predicted and actual values. A lower MSE indicates better performance of the model.

We will also use MAE (Mean Absolute Error) as an additional metric to evaluate the model's performance. MAE measures the average absolute difference between the predicted and actual values, providing a more interpretable measure of prediction accuracy.

Data Dictionary

Each row in this file corresponds to one unique delivery. Each column corresponds to a feature as explained below.

1. market_id : integer id for the market where the restaurant lies
2. created_at : the timestamp at which the order was placed
3. actual_delivery_time : the timestamp when the order was delivered
4. store_primary_category : category for the restaurant
5. order_protocol : integer code value for order protocol(how the order was placed ie: through porter, call to restaurant, pre booked, third part etc)
6. total_items subtotal : final price of the order
7. num_distinct_items : the number of distinct items in the order
8. min_item_price : price of the cheapest item in the order
9. max_item_price : price of the costliest item in order
10. total_onshift_partners : number of delivery partners on duty at the time order was placed
11. total_busy_partners : number of delivery partners attending to other tasks
12. total_outstanding_orders : total number of orders to be fulfilled at the moment

In [1]:
```
import polars as pl
import duckdb as db
import numpy as np
from sklearn.impute import KNNImputer
from sklearn.preprocessing import StandardScaler
```

In [2]:
```
df = pl.read_csv("../data/raw/dataset.csv")
```

Data Exploration

In [3]:
```
df.head()
```

Out[3]: shape: (5, 14)

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshif
f64	str	str	str	str	f64	i64	i64	i64	i64	i64	
1.0	"2015-02-06 22:24:17"	"2015-02-06 23:27:16"	"df263d996281d984952c07998dc543...	"american"	1.0	4	3441	4	557	1239	
2.0	"2015-02-10 21:49:25"	"2015-02-10 22:56:29"	"f0ade77b43923b38237db569b016ba...	"mexican"	2.0	1	1900	1	1400	1400	
3.0	"2015-01-22 20:39:28"	"2015-01-22 21:09:09"	"f0ade77b43923b38237db569b016ba...	null	1.0	1	1900	1	1900	1900	
3.0	"2015-02-03 21:21:45"	"2015-02-03 22:13:00"	"f0ade77b43923b38237db569b016ba...	null	1.0	6	6900	5	600	1800	
3.0	"2015-02-15 02:40:36"	"2015-02-15 03:20:26"	"f0ade77b43923b38237db569b016ba...	null	1.0	3	3900	3	1100	1600	

In [4]:
```
df.shape
```

Out[4]: (197428, 14)

In [5]:
```
df.describe()
```

Out[5]: shape: (9, 15)

statistic	market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_pri
str	f64	str	str	str	str	f64	f64	f64	f64	f64	f
"count"	196441.0	"197428"	"197421"	"197428"	"192668"	196433.0	197428.0	197428.0	197428.0	197428.0	197428
"null_count"	987.0	"0"	"7"	"0"	"4760"	995.0	0.0	0.0	0.0	0.0	0
"mean"	2.978706	null	null	null	null	2.882352	3.196391	2682.331402	2.670791	686.21847	1159.588
"std"	1.524867	null	null	null	null	1.503771	2.666546	1823.093688	1.630255	522.038648	558.4113
"min"	1.0	"2014-10-19 05:24:15"	"2015-01-21 15:58:11"	"0004d0b59e19461ff126e3a08a814c...	"afghan"	1.0	1.0	0.0	1.0	-86.0	0
"25%"	2.0	null	null	null	null	1.0	2.0	1400.0	1.0	299.0	800
"50%"	3.0	null	null	null	null	3.0	3.0	2200.0	2.0	595.0	1095
"75%"	4.0	null	null	null	null	4.0	4.0	3395.0	3.0	949.0	1395
"max"	6.0	"2015-02-18 06:00:44"	"2015-02-19 22:45:31"	"ffedf5be3a86e2ee281d54cdc97bc1...	"vietnamese"	7.0	411.0	27100.0	20.0	14700.0	14700

In [6]:
```
dict(zip(df.columns, df.dtypes))
```

Out[6]:
```
{'market_id': Float64,
 'created_at': String,
 'actual_delivery_time': String,
 'store_id': String,
 'store_primary_category': String,
 'order_protocol': Float64,
 'total_items': Int64,
 'subtotal': Int64,
 'num_distinct_items': Int64,
 'min_item_price': Int64,
 'max_item_price': Int64,
 'total_onshift_partners': Float64,
 'total_busy_partners': Float64,
 'total_outstanding_orders': Float64}
```

In [7]:
```
df.null_count()
```

Out[7]: shape: (1, 14)

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshift_partners	total_busy_partne
u32	u32	u32	u32	u32	u32	u32	u32	u32	u32	u32	u32	u
987	0	7	0	4760	995	0	0	0	0	0	16262	162

In [8]:
```
db.sql("""
select * from df where total_onshift_partners IS NULL limit 20
""")
```

Out[8]:

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshift_partners	total_busy_partners	total_outstanding_orders
double	varchar	varchar	varchar	varchar	double	int64	int64	int64	int64	int64	double	double	double
6.0	2015-02-06 01:11:56	2015-02-06 01:42:51	45d38ce7f5231602e24a2103a0300ae6	breakfast	2.0	2	575	2	225	350	NULL	NULL	NULL
6.0	2015-02-14 02:07:47	2015-02-14 03:17:37	45d38ce7f5231602e24a2103a0300ae6	breakfast	2.0	5	1415	3	185	675	NULL	NULL	NULL
6.0	2015-01-31 21:58:30	2015-01-31 22:55:32	45d38ce7f5231602e24a2103a0300ae6	breakfast	2.0	1	650	1	650	650	NULL	NULL	NULL
6.0	2015-02-08 03:28:59	2015-02-08 05:32:11	45d38ce7f5231602e24a2103a0300ae6	breakfast	2.0	5	1550	5	225	700	NULL	NULL	NULL
6.0	2015-01-23 19:29:17	2015-01-23 20:25:25	45d38ce7f5231602e24a2103a0300ae6	breakfast	2.0	6	1110	5	185	185	NULL	NULL	NULL
6.0	2015-02-03 22:27:24	2015-02-03 23:09:31	45d38ce7f5231602e24a2103a0300ae6	breakfast	2.0	8	2115	8	150	650	NULL	NULL	NULL
6.0	2015-01-31 01:35:23	2015-01-31 02:16:43	45d38ce7f5231602e24a2103a0300ae6	breakfast	2.0	2	1741	2	889	772	NULL	NULL	NULL
6.0	2015-01-24 23:17:11	2015-01-25 00:24:49	45d38ce7f5231602e24a2103a0300ae6	breakfast	2.0	7	2250	2	225	900	NULL	NULL	NULL
6.0	2015-02-03 21:15:23	2015-02-03 22:38:32	45d38ce7f5231602e24a2103a0300ae6	breakfast	2.0	1	3900	1	3900	3900	NULL	NULL	NULL
6.0	2015-02-15 19:57:36	2015-02-15 20:49:04	4a06d868d044c50af0cf9bc82d2fc19f	sandwich	1.0	4	3575	4	400	925	NULL	NULL	NULL
6.0	2015-01-25 19:15:56	2015-01-25 19:40:38	4a06d868d044c50af0cf9bc82d2fc19f	sandwich	1.0	2	2374	2	979	1048	NULL	NULL	NULL
6.0	2015-01-28 19:50:12	2015-01-28 20:41:22	e143c01e314f7b950daca31188cb5d0f	sandwich	1.0	7	2503	4	0	1199	NULL	NULL	NULL
6.0	2015-01-28 21:34:15	2015-01-28 22:01:36	e143c01e314f7b950daca31188cb5d0f	sandwich	1.0	5	1997	5	179	649	NULL	NULL	NULL
6.0	2015-02-17 19:38:04	2015-02-17 20:02:27	e143c01e314f7b950daca31188cb5d0f	sandwich	1.0	3	1274	3	0	375	NULL	NULL	NULL
6.0	2015-01-29 21:18:12	2015-01-29 21:43:05	e143c01e314f7b950daca31188cb5d0f	sandwich	1.0	2	1798	1	899	899	NULL	NULL	NULL
6.0	2015-02-14 21:20:04	2015-02-14 21:54:10	e143c01e314f7b950daca31188cb5d0f	sandwich	1.0	2	2497	2	999	1199	NULL	NULL	NULL
6.0	2015-02-03 23:33:16	2015-02-04 00:07:15	e143c01e314f7b950daca31188cb5d0f	sandwich	4.0	2	1597	2	299	1099	NULL	NULL	NULL
6.0	2015-02-07 03:40:27	2015-02-07 04:06:48	e143c01e314f7b950daca31188cb5d0f	sandwich	1.0	1	1048	1	699	699	NULL	NULL	NULL
6.0	2015-02-07 21:28:56	2015-02-07 22:00:43	e143c01e314f7b950daca31188cb5d0f	sandwich	1.0	2	1199	2	300	899	NULL	NULL	NULL
6.0	2015-02-09 05:22:41	2015-02-09 05:51:35	1f0643149dd5c86751215fd0040675aa	korean	1.0	2	2390	2	1095	1295	NULL	NULL	NULL

20 rows
14 columns

In [9]:
```
db.sql("""
select * from df where store_primary_category IS NULL limit 20
""")
```

Out[9]:

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshift_partners	total_busy_partners	total_outstanding_orders
double	varchar	varchar	varchar	varchar	double	int64	int64	int64	int64	int64	double	double	double
3.0	2015-01-22 20:39:28	2015-01-22 21:09:09	f0ade77b43923b38237db569b016ba25	NULL	1.0	1	1900	1	1900	1900	1.0	0.0	0.0
3.0	2015-02-03 21:21:45	2015-02-03 22:13:00	f0ade77b43923b38237db569b016ba25	NULL	1.0	6	6900	5	600	1800	1.0	1.0	2.0
3.0	2015-02-15 02:40:36	2015-02-15 03:20:26	f0ade77b43923b38237db569b016ba25	NULL	1.0	3	3900	3	1100	1600	6.0	6.0	9.0
3.0	2015-01-28 20:30:38	2015-01-28 21:08:58	f0ade77b43923b38237db569b016ba25	NULL	1.0	3	5000	3	1500	1900	2.0	2.0	2.0
3.0	2015-01-31 02:16:36	2015-01-31 02:43:00	f0ade77b43923b38237db569b016ba25	NULL	1.0	2	3900	2	1200	2700	10.0	9.0	9.0
3.0	2015-02-12 03:03:35	2015-02-12 03:36:20	f0ade77b43923b38237db569b016ba25	NULL	1.0	4	4850	4	750	1800	7.0	8.0	7.0
3.0	2015-02-18 01:15:45	2015-02-18 02:08:57	f0ade77b43923b38237db569b016ba25	NULL	1.0	2	2100	2	700	1200	2.0	2.0	2.0
3.0	2015-02-02 19:22:53	2015-02-02 20:09:19	f0ade77b43923b38237db569b016ba25	NULL	4.0	4	4300	4	1200	1500	1.0	1.0	1.0
3.0	2015-02-16 04:19:33	2015-02-16 06:34:00	f0ade77b43923b38237db569b016ba25	NULL	1.0	2	2200	2	600	1600	3.0	3.0	4.0
3.0	2015-02-07 01:34:31	2015-02-07 02:17:14	f0ade77b43923b38237db569b016ba25	NULL	1.0	1	1900	1	1900	1900	6.0	3.0	3.0
3.0	2015-01-25 01:50:51	2015-01-25 02:28:53	f0ade77b43923b38237db569b016ba25	NULL	4.0	4	4986	4	699	2362	16.0	6.0	9.0
1.0	2015-01-28 20:33:04	2015-01-28 21:04:14	50905d7b2216bfeccb5b41016357176b	NULL	NULL	3	1765	3	275	895	22.0	21.0	24.0
1.0	2015-02-12 20:27:42	2015-02-12 21:18:29	50905d7b2216bfeccb5b41016357176b	NULL	2.0	1	975	1	975	975	24.0	25.0	30.0
1.0	2015-01-28 21:37:38	2015-01-28 22:11:53	50905d7b2216bfeccb5b41016357176b	NULL	2.0	3	1425	3	175	900	13.0	12.0	7.0
1.0	2015-01-21 20:35:27	2015-01-21 21:01:26	50905d7b2216bfeccb5b41016357176b	NULL	2.0	2	1750	2	825	825	18.0	18.0	20.0
1.0	2015-02-16 19:19:17	2015-02-16 20:16:43	50905d7b2216bfeccb5b41016357176b	NULL	2.0	8	7870	6	825	895	12.0	11.0	9.0
1.0	2015-01-27 20:26:19	2015-01-27 21:03:53	50905d7b2216bfeccb5b41016357176b	NULL	2.0	2	1725	2	825	900	21.0	18.0	18.0
1.0	2015-01-22 20:19:08	2015-01-22 21:07:43	50905d7b2216bfeccb5b41016357176b	NULL	2.0	3	2015	3	395	1025	22.0	13.0	13.0
1.0	2015-02-10 20:05:17	2015-02-10 20:34:43	50905d7b2216bfeccb5b41016357176b	NULL	2.0	1	1220	1	825	825	21.0	21.0	20.0
3.0	2015-01-23 19:22:18	2015-01-23 19:48:50	51b7dae1031b20174cacc7e69d6e4bf0	NULL	4.0	3	582	1	194	194	6.0	5.0	4.0

20 rows
14 columns

In [10]:
```
db.sql("""
select * from df where order_protocol IS NULL limit 20
""")
```

Out[10]:

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshift_partners	total_busy_partners	total_outstanding_orders
double	varchar	varchar	varchar	varchar	double	int64	int64	int64	int64	int64	double	double	double
1.0	2015-01-28 20:33:04	2015-01-28 21:04:14	50905d7b2216bfeccb5b41016357176b	NULL	NULL	3	1765	3	275	895	22.0	21.0	24.0
4.0	2015-01-24 19:48:45	2015-01-24 20:31:14	a87ff679a2f3e71d9181a67b7542122c	mediterranean	NULL	2	2070	2	325	1745	36.0	60.0	59.0
2.0	2015-02-10 19:36:38	2015-02-10 20:24:14	26324d8e2cc1957b8e581568a089a51c	NULL	NULL	9	4781	8	379	779	37.0	38.0	68.0
NULL	2015-02-17 02:17:43	2015-02-17 03:15:14	fe8c15fed5f808006ce95eddb7366e35	NULL	NULL	3	3400	3	1000	1200	36.0	36.0	52.0
4.0	2015-02-09 02:10:29	2015-02-09 02:59:20	6ea9ab1baa0efb9e19094440c317e21b	NULL	NULL	2	2790	2	1295	1495	84.0	66.0	121.0
5.0	2015-01-30 20:27:37	2015-01-30 21:17:51	b597976c3ce6012f3a07e9f5c71a3c8c	NULL	NULL	1	1995	1	1995	1995	14.0	10.0	12.0
3.0	2015-01-23 02:21:51	2015-01-23 03:29:52	ed71773d43d53fa70ecf593c6582d9cc	sushi	NULL	6	5275	4	400	1295	8.0	7.0	7.0
NULL	2015-02-17 03:49:46	2015-02-17 04:21:27	86311dbe35f1b6c5166365165602f54d	pizza	NULL	1	1699	1	1399	1399	44.0	40.0	69.0
NULL	2015-02-07 02:11:12	2015-02-07 03:10:11	29cb2d8a436d82bf5dbcca7ea09edc5c	NULL	NULL	5	5240	5	595	1325	47.0	40.0	71.0
NULL	2015-01-23 04:09:03	2015-01-23 05:14:00	9f8684e630c4c30cad7b1f0935cd62ab	NULL	NULL	2	2240	1	920	920	NULL	NULL	NULL
4.0	2015-02-10 23:44:10	2015-02-11 00:34:39	cf2226ddd41b1a2d0ae51dab54d32c36	fast	NULL	2	1850	1	775	775	11.0	12.0	21.0
4.0	2015-01-29 01:09:59	2015-01-29 02:01:41	cf2226ddd41b1a2d0ae51dab54d32c36	NULL	NULL	3	1967	3	189	659	81.0	70.0	82.0
NULL	2015-01-29 02:46:37	2015-01-29 03:47:41	fd8c07a31f8a85910ad8476f5f7efb27	chinese	NULL	2	890	2	160	710	0.0	0.0	0.0
2.0	2015-02-16 02:05:20	2015-02-16 02:44:08	0bb0846327772451045bd30dd347821b	NULL	NULL	1	2575	1	2575	2575	55.0	54.0	89.0
1.0	2015-02-07 03:42:08	2015-02-07 05:00:30	35b47299c1130953d286c976e9c608dc	british	NULL	2	1750	2	200	1550	23.0	25.0	38.0
NULL	2015-01-21 20:42:48	2015-01-21 21:20:59	58e4d44e550d0f7ee0a23d6b02d9b0db	american	NULL	3	2150	2	500	1150	24.0	24.0	28.0
5.0	2015-02-15 18:24:36	2015-02-15 18:50:49	bb838bff2c2120b9237cac812c1019f8	NULL	NULL	6	1670	4	175	395	6.0	6.0	6.0
3.0	2015-02-07 23:05:03	2015-02-07 23:35:05	8de0c3085da54b8e957220b9c8de8aca	NULL	NULL	6	3538	3	100	1050	7.0	8.0	13.0
2.0	2015-01-24 05:14:44	2015-01-24 05:55:35	34be90676412d14c80f85809d8e3b97b	NULL	NULL	4	4696	2	1099	1199	27.0	27.0	47.0
1.0	2015-01-29 00:57:22	2015-01-29 01:40:19	f187a23c3ee681ef6913f31fd6d6446b	pizza	NULL	1	995	1	695	695	11.0	7.0	6.0

20 rows
14 columns

In [11]:
```
db.sql("""
select market_id, store_primary_category, count(*) from df group by market_id, store_primary_category order by count(*) desc limit 20
""")
```

Out[11]:

market_id double	store_primary_category varchar	count_star() int64
2.0	mexican	6647
2.0	american	5700
1.0	american	5228
2.0	pizza	4859
1.0	pizza	4081
4.0	pizza	3840
2.0	sandwich	3751
2.0	burger	3340
4.0	mexican	3250
2.0	dessert	3115
4.0	chinese	2847
3.0	american	2743
1.0	japanese	2628
4.0	burger	2603
4.0	dessert	2601
4.0	vietnamese	2541
3.0	mexican	2487
1.0	mexican	2439
4.0	japanese	2403
5.0	american	2386
20 rows		3 columns

In [12]:
```
df["store_primary_category"].value_counts().sort("count", descending=True)
```

Out[12]: shape: (75, 2)

store_primary_category	count
str	u32
"american"	19399
"pizza"	17321
"mexican"	17099
"burger"	10958
"sandwich"	10060
...	...
"lebanese"	9
"belgian"	2
"indonesian"	2
"alcohol-plus-food"	1
"chocolate"	1

In [13]:
```
df["order_protocol"].value_counts().sort("count", descending=True)
```

Out[13]: shape: (8, 2)

order_protocol	count
f64	u32
1.0	54725
3.0	53199
5.0	44290
2.0	24052
4.0	19354
null	995
6.0	794
7.0	19

In [14]:
```
df["market_id"].value_counts().sort("count", descending=True)
```

Out[14]: shape: (7, 2)

market_id	count
f64	u32
2.0	55058
4.0	47599
1.0	38037
3.0	23297
5.0	18000
6.0	14450
null	987

In [15]:
```
df=df.with_columns(
    pl.col("min_item_price").abs(),
    pl.col("max_item_price").abs(),
    pl.col("total_onshift_partners").abs(),
    pl.col("total_busy_partners").abs(),
    pl.col("total_outstanding_orders").abs(),
)
```

In [16]:
```
df.describe()
```

Out[16]: shape: (9, 15)

statistic	market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_pri
str	f64	str	str	str	str	f64	f64	f64	f64	f64	f
"count"	196441.0	"197428"	"197421"	"197428"	"192668"	196433.0	197428.0	197428.0	197428.0	197428.0	197428
"null_count"	987.0	"0"	"7"	"0"	"4760"	995.0	0.0	0.0	0.0	0.0	0
"mean"	2.978706	null	null	null	null	2.882352	3.196391	2682.331402	2.670791	686.222339	1159.588
"std"	1.524867	null	null	null	null	1.503771	2.666546	1823.093688	1.630255	522.033561	558.4113
"min"	1.0	"2014-10-19 05:24:15"	"2015-01-21 15:58:11"	"0004d0b59e19461ff126e3a08a814c...	"afghan"	1.0	1.0	0.0	1.0	0.0	0
"25%"	2.0	null	null	null	null	1.0	2.0	1400.0	1.0	299.0	800
"50%"	3.0	null	null	null	null	3.0	3.0	2200.0	2.0	595.0	1095
"75%"	4.0	null	null	null	null	4.0	4.0	3395.0	3.0	949.0	1395
"max"	6.0	"2015-02-18 06:00:44"	"2015-02-19 22:45:31"	"ffedf5be3a86e2ee281d54cdc97bc1...	"vietnamese"	7.0	411.0	27100.0	20.0	14700.0	14700

In [17]:
```
df.null_count()
```

Out[17]: shape: (1, 14)

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshift_partners	total_busy_partne
u32	u32	u32	u32	u32	u32	u32	u32	u32	u32	u32	u32	u
987	0	7	0	4760	995	0	0	0	0	0	16262	162

Null Values Imputation

In [18]:
```
df = df.drop_nulls(subset=["total_onshift_partners", "total_busy_partners", "total_outstanding_orders", "actual_delivery_time",\
                        #    "market_id","store_primary_category","order_protocol"\
                           ])
```

In [19]:
```
df.describe()
```

Out[19]: shape: (9, 15)

statistic	market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_pr
str	f64	str	str	str	str	f64	f64	f64	f64	f64	f
"count"	180240.0	"181159"	"181159"	"181159"	"176944"	180242.0	181159.0	181159.0	181159.0	181159.0	18115
"null_count"	919.0	"0"	"0"	"0"	"4215"	917.0	0.0	0.0	0.0	0.0	
"mean"	2.748702	null	null	null	null	2.895923	3.208469	2698.524296	2.677659	684.854095	1160.7526
"std"	1.331715	null	null	null	null	1.514983	2.673052	1829.304441	1.627291	521.286542	561.8386
"min"	1.0	"2015-01-21 15:22:03"	"2015-01-21 15:58:11"	"0004d0b59e19461ff126e3a08a814c..."	"afghan"	1.0	1.0	0.0	1.0	0.0	
"25%"	2.0	null	null	null	null	1.0	2.0	1415.0	2.0	299.0	79
"50%"	2.0	null	null	null	null	3.0	3.0	2225.0	2.0	595.0	109
"75%"	4.0	null	null	null	null	4.0	4.0	3411.0	3.0	942.0	139
"max"	6.0	"2015-02-18 06:00:44"	"2015-02-19 22:45:31"	"ffeabd223de0d4eacb9a3e6e53e544..."	"vietnamese"	7.0	411.0	26800.0	20.0	14700.0	1470

In [20]:
```
db.sql("""
    select * from df where max_item_price = 0 and min_item_price = 0
""")
```

Out[20]:
```
┌────────────┬─────────────────────┬──────────────────────┬──────────────────────────────────┬──────────────────────┬──────────────────┬─────────────┬──────────┬──────────────────┬──────────────┬────────────────┐
│ market_id  │     created_at      │ actual_delivery_time │             store_id             │ store_primary_category │ order_protocol │ total_items │ subtotal │ num_distinct_items │ min_item_price │ max_item_price │
│   double   │       varchar       │       varchar        │             varchar              │        varchar         │      double    │    int64    │  int64   │       int64        │     int64      │     int64      │
├────────────┼─────────────────────┼──────────────────────┼──────────────────────────────────┼──────────────────────┼──────────────────┼─────────────┼──────────┼──────────────────┼──────────────┼────────────────┤
│        1.0 │ 2015-02-09 18:23:31 │  2015-02-09 19:31:54 │ 3e91970f771a2c473ae36b60d1146068 │ mexican              │              1.0 │           1 │      575 │                1 │            0 │              0 │
│        1.0 │ 2015-01-22 17:57:08 │  2015-01-22 18:54:37 │ 3e91970f771a2c473ae36b60d1146068 │ mexican              │              1.0 │           1 │      575 │                1 │            0 │              0 │
│        1.0 │ 2015-02-13 18:41:31 │  2015-02-13 19:30:39 │ 3e91970f771a2c473ae36b60d1146068 │ mexican              │              1.0 │           1 │      575 │                1 │            0 │              0 │
│        1.0 │ 2015-01-21 19:36:57 │  2015-01-21 20:23:45 │ 3e91970f771a2c473ae36b60d1146068 │ mexican              │              1.0 │           1 │      560 │                1 │            0 │              0 │
│        1.0 │ 2015-02-13 02:37:10 │  2015-02-13 03:13:44 │ 3e91970f771a2c473ae36b60d1146068 │ dim-sum              │              5.0 │           1 │      595 │                1 │            0 │              0 │
│        5.0 │ 2015-01-27 02:26:45 │  2015-01-27 02:56:56 │ 648f4baa45889f9c5f4f7add35862841 │ pizza                │              3.0 │           1 │     2418 │                1 │            0 │              0 │
│        1.0 │ 2015-01-21 19:50:42 │  2015-01-21 20:24:07 │ da5e8bfed9bdb84595be92afeb3fd378 │ sandwich             │              1.0 │           2 │     1090 │                2 │            0 │              0 │
└────────────┴─────────────────────┴──────────────────────┴──────────────────────────────────┴──────────────────────┴──────────────────┴─────────────┴──────────┴──────────────────┴──────────────┴────────────────┘
```

In [21]:
```
df=df.with_columns(
    pl.when(pl.col("max_item_price") == 0).then(pl.col("max_item_price").mean()).otherwise(pl.col("max_item_price")).alias("max_item_price"),
    pl.when(pl.col("min_item_price") == 0).then(pl.col("min_item_price").mean()).otherwise(pl.col("min_item_price")).alias("min_item_price")
)
```

In [22]:
```
db.sql("""
    select * from df where max_item_price < min_item_price
""")
```

Out[22]:

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshift_partners	total_busy_partners	total_outstanding_orders
double	varchar	varchar	varchar	varchar	double	int64	int64	int64	double	double	double	double	double
4.0	2015-01-24 03:03:18	2015-01-24 04:02:30	a87ff679a2f3e71d9181a67b7542122c	mexican	1.0	1	1403	1	1314.0	1292.0	129.0	127.0	205.0
1.0	2015-02-03 03:36:16	2015-02-03 05:13:04	c56a4706337730e0e15da875405fa1c5	fast	4.0	7	1223	5	684.8540950215005	649.0	22.0	22.0	48.0
1.0	2015-01-25 21:38:45	2015-01-25 22:14:22	c56a4706337730e0e15da875405fa1c5	fast	4.0	6	958	6	684.8540950215005	599.0	17.0	9.0	9.0
5.0	2015-01-31 20:10:47	2015-01-31 20:49:29	b597976c3ce6012f3a07e9f5c71a3c8c	NULL	1.0	1	1289	1	1437.0	1400.0	21.0	13.0	19.0
5.0	2015-01-30 02:20:58	2015-01-30 03:14:42	b597976c3ce6012f3a07e9f5c71a3c8c	NULL	1.0	1	1828	1	2031.0	1935.0	24.0	21.0	23.0
2.0	2015-01-27 04:09:54	2015-01-27 04:56:33	f806c5d2707545d718717be03e69a8d4	fast	4.0	14	1305	3	684.8540950215005	190.0	65.0	93.0	77.0
2.0	2015-02-03 03:22:21	2015-02-03 04:14:00	f806c5d2707545d718717be03e69a8d4	fast	4.0	13	985	4	684.8540950215005	390.0	96.0	89.0	154.0
5.0	2015-02-08 03:42:21	2015-02-08 04:20:11	f806c5d2707545d718717be03e69a8d4	nepalese	1.0	31	2820	7	684.8540950215005	355.0	116.0	120.0	184.0
2.0	2015-01-31 19:28:49	2015-01-31 20:17:29	f806c5d2707545d718717be03e69a8d4	fast	4.0	40	3000	2	684.8540950215005	150.0	48.0	45.0	51.0
2.0	2015-02-07 05:47:42	2015-02-07 06:26:49	f806c5d2707545d718717be03e69a8d4	fast	4.0	26	2320	5	684.8540950215005	250.0	39.0	43.0	67.0
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3.0	2015-02-06 20:06:59	2015-02-06 20:40:50	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	1	1299	1	805.0	765.0	6.0	6.0	9.0
4.0	2015-02-03 20:36:37	2015-02-03 21:04:49	4cc39b42f22ee5987819b83fc055cb33	pizza	3.0	1	625	1	553.0	540.0	31.0	35.0	31.0
4.0	2015-02-07 20:52:15	2015-02-07 21:28:58	4cc39b42f22ee5987819b83fc055cb33	pizza	3.0	1	3174	1	1513.0	1403.0	38.0	31.0	32.0
4.0	2015-02-16 02:34:06	2015-02-16 03:20:42	17e62166fc8586dfa4d1bc0e1742c08b	vietnamese	5.0	2	2086	1	904.0	878.0	85.0	77.0	152.0
4.0	2015-02-03 02:57:44	2015-02-03 03:41:32	17e62166fc8586dfa4d1bc0e1742c08b	vietnamese	5.0	2	2082	2	1020.0	928.0	88.0	87.0	150.0
1.0	2015-02-15 22:15:36	2015-02-15 22:42:14	1763ea5a7e72dd7ee64073c2dda7a7a8	fast	4.0	9	946	5	684.8540950215005	299.0	24.0	17.0	17.0
1.0	2015-02-09 05:16:26	2015-02-09 05:48:49	1763ea5a7e72dd7ee64073c2dda7a7a8	fast	4.0	6	348	3	684.8540950215005	169.0	20.0	20.0	18.0
1.0	2015-01-25 03:06:25	2015-01-25 03:32:31	1763ea5a7e72dd7ee64073c2dda7a7a8	fast	4.0	11	1351	5	684.8540950215005	169.0	75.0	66.0	83.0
1.0	2015-01-30 19:24:55	2015-01-30 20:03:21	a914ecef9c12ffdb9bede64bb703d877	fast	4.0	1	1142	1	737.0	701.0	22.0	19.0	32.0
1.0	2015-02-09 20:32:27	2015-02-09 21:09:10	a914ecef9c12ffdb9bede64bb703d877	fast	4.0	1	437	1	497.0	448.0	36.0	26.0	27.0

1778 rows (20 shown)
14 columns

In [23]:
```
df=df.with_columns(
    max_price_temp=pl.col('max_item_price'),
    min_price_temp=pl.col('min_item_price'),
)
```

In [24]:
```
df=df.with_columns(
    max_item_price=pl.when(pl.col('max_price_temp') < pl.col('min_price_temp'))
    .then(pl.col('min_price_temp'))
    .otherwise(pl.col('max_price_temp')),
    min_item_price=pl.when(pl.col('min_price_temp') > pl.col('max_price_temp'))
    .then(pl.col('max_price_temp'))
    .otherwise(pl.col('min_price_temp'))
).drop('max_price_temp', 'min_price_temp')
```

In [25]:
```
db.sql("""
    select * from df where  total_onshift_partners = 0
""")
```

Out[25]:

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshift_partners	total_busy_partners	total_outstanding_orders
double	varchar	varchar	varchar	varchar	double	int64	int64	int64	double	double	double	double	double
1.0	2015-01-27 02:29:00	2015-01-27 03:30:45	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	1	1150	1	1150.0	1150.0	0.0	0.0	0.0
1.0	2015-02-08 00:56:26	2015-02-08 01:50:48	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	4	4805	2	1095.0	1195.0	0.0	0.0	0.0
1.0	2015-01-24 04:01:10	2015-01-24 04:28:57	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	7	5200	5	225.0	1095.0	0.0	0.0	0.0
1.0	2015-02-01 06:20:26	2015-02-01 06:55:26	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	3	2910	3	595.0	1095.0	0.0	0.0	0.0
1.0	2015-02-14 03:10:56	2015-02-14 04:12:41	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	2	2495	2	1095.0	1150.0	0.0	0.0	0.0
1.0	2015-02-04 19:28:18	2015-02-04 20:11:13	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	2	2345	2	1150.0	1195.0	0.0	0.0	0.0
1.0	2015-01-25 03:22:11	2015-01-25 04:14:22	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	1	1250	1	1150.0	1150.0	0.0	0.0	0.0
1.0	2015-02-08 23:06:13	2015-02-08 23:42:58	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	2	1190	2	595.0	595.0	0.0	0.0	0.0
1.0	2015-01-23 02:45:55	2015-01-23 03:32:26	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	7	5035	5	195.0	1150.0	0.0	0.0	0.0
1.0	2015-01-23 05:03:26	2015-01-23 05:56:30	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	3	2065	3	195.0	1150.0	0.0	0.0	0.0
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3.0	2015-01-28 16:18:30	2015-01-28 17:33:30	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	2	1640	2	615.0	925.0	0.0	0.0	0.0
3.0	2015-02-17 19:03:53	2015-02-17 19:51:14	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	3	2500	3	500.0	975.0	0.0	0.0	0.0
3.0	2015-02-14 17:31:55	2015-02-14 18:24:49	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	3	2700	3	650.0	975.0	0.0	0.0	0.0
3.0	2015-02-10 19:03:48	2015-02-10 20:26:52	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	3	1415	3	245.0	810.0	0.0	0.0	0.0
3.0	2015-02-12 21:07:28	2015-02-12 21:55:41	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	3	1415	3	245.0	810.0	0.0	0.0	0.0
3.0	2015-01-23 17:53:47	2015-01-23 18:42:14	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	1	975	1	975.0	975.0	0.0	0.0	0.0
3.0	2015-02-07 15:31:33	2015-02-07 16:24:02	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	5	2430	4	270.0	790.0	0.0	0.0	0.0
3.0	2015-02-08 16:11:00	2015-02-08 17:36:23	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	4	2960	4	480.0	925.0	0.0	0.0	0.0
3.0	2015-01-21 17:13:16	2015-01-21 17:51:09	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	2	1175	2	345.0	790.0	0.0	0.0	0.0
3.0	2015-01-24 05:56:06	2015-01-24 06:27:21	10e4d7889812f78893b86aeb04111871	mexican	3.0	1	2599	1	2599.0	2599.0	0.0	0.0	2.0

3615 rows (20 shown)
14 columns

In [26]:
```
db.sql("""
    select * from df where total_busy_partners = 0 and total_onshift_partners = 0 and total_outstanding_orders = 0
""")
```

Out[26]:

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshift_partners	total_busy_partners	total_outstanding_orders
double	varchar	varchar	varchar	varchar	double	int64	int64	int64	double	double	double	double	double
1.0	2015-01-27 02:29:00	2015-01-27 03:30:45	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	1	1150	1	1150.0	1150.0	0.0	0.0	0.0
1.0	2015-02-08 00:56:26	2015-02-08 01:50:48	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	4	4805	2	1095.0	1195.0	0.0	0.0	0.0
1.0	2015-01-24 04:01:10	2015-01-24 04:28:57	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	7	5200	5	225.0	1095.0	0.0	0.0	0.0
1.0	2015-02-01 06:20:26	2015-02-01 06:55:26	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	3	2910	3	595.0	1095.0	0.0	0.0	0.0
1.0	2015-02-14 03:10:56	2015-02-14 04:12:41	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	2	2495	2	1095.0	1150.0	0.0	0.0	0.0
1.0	2015-02-04 19:28:18	2015-02-04 20:11:13	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	2	2345	2	1150.0	1195.0	0.0	0.0	0.0
1.0	2015-01-25 03:22:11	2015-01-25 04:14:22	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	1	1250	1	1150.0	1150.0	0.0	0.0	0.0
1.0	2015-02-08 23:06:13	2015-02-08 23:42:58	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	2	1190	2	595.0	595.0	0.0	0.0	0.0
1.0	2015-01-23 02:45:55	2015-01-23 03:32:26	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	7	5035	5	195.0	1150.0	0.0	0.0	0.0
1.0	2015-01-23 05:03:26	2015-01-23 05:56:30	5a01f0597ac4bdf35c24846734ee9a76	japanese	1.0	3	2065	3	195.0	1150.0	0.0	0.0	0.0
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3.0	2015-01-24 17:11:50	2015-01-24 18:29:10	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	3	2455	3	480.0	950.0	0.0	0.0	0.0
3.0	2015-01-28 16:18:30	2015-01-28 17:33:30	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	2	1640	2	615.0	925.0	0.0	0.0	0.0
3.0	2015-02-17 19:03:53	2015-02-17 19:51:14	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	3	2500	3	500.0	975.0	0.0	0.0	0.0
3.0	2015-02-14 17:31:55	2015-02-14 18:24:49	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	3	2700	3	650.0	975.0	0.0	0.0	0.0
3.0	2015-02-10 19:03:48	2015-02-10 20:26:52	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	3	1415	3	245.0	810.0	0.0	0.0	0.0
3.0	2015-02-12 21:07:28	2015-02-12 21:55:41	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	3	1415	3	245.0	810.0	0.0	0.0	0.0
3.0	2015-01-23 17:53:47	2015-01-23 18:42:14	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	1	975	1	975.0	975.0	0.0	0.0	0.0
3.0	2015-02-07 15:31:33	2015-02-07 16:24:02	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	5	2430	4	270.0	790.0	0.0	0.0	0.0
3.0	2015-02-08 16:11:00	2015-02-08 17:36:23	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	4	2960	4	480.0	925.0	0.0	0.0	0.0
3.0	2015-01-21 17:13:16	2015-01-21 17:51:09	dfb61b74af460c2fd68bb8266f9f0814	breakfast	1.0	2	1175	2	345.0	790.0	0.0	0.0	0.0

3415 rows (20 shown)
14 columns

In [27]:
```
df=df.filter(
    ~(pl.col("total_onshift_partners") == 0)
)
```

In [28]:
```
df.describe()
```

Out[28]: shape: (9, 15)

statistic	market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_pri
str	f64	str	str	str	str	f64	f64	f64	f64	f64	f6
"count"	176642.0	"177544"	"177544"	"177544"	"173383"	176640.0	177544.0	177544.0	177544.0	177544.0	177544
"null_count"	902.0	"0"	"0"	"0"	"4161"	904.0	0.0	0.0	0.0	0.0	0
"mean"	2.767864	null	null	null	null	2.898687	3.205588	2701.713919	2.675618	692.093454	1164.9864
"std"	1.329898	null	null	null	null	1.513229	2.67464	1829.211488	1.62441	516.496735	559.2193
"min"	1.0	"2015-01-21 15:22:03"	"2015-01-21 15:58:11"	"0004d0b59e19461ff126e3a08a814c...	"afghan"	1.0	1.0	0.0	1.0	1.0	52
"25%"	2.0	null	null	null	null	1.0	2.0	1420.0	2.0	300.0	799
"50%"	2.0	null	null	null	null	3.0	3.0	2228.0	2.0	599.0	1095
"75%"	4.0	null	null	null	null	4.0	4.0	3421.0	3.0	943.0	1397
"max"	6.0	"2015-02-18 06:00:44"	"2015-02-19 22:45:31"	"ffeabd223de0d4eacb9a3e6e53e544...	"vietnamese"	7.0	411.0	26800.0	20.0	14700.0	14700

In [29]:
```
category_per_store = df.group_by("store_id").agg(
    pl.col("store_primary_category").mode().first().alias("store_primary_category_name")
)
category_per_store = category_per_store.fill_null("other")
```

In [30]:
```
df = df.join(category_per_store, on="store_id", how="left")
```

In [31]:
```
df = df.with_columns(
   pl.when(pl.col("store_primary_category").is_null()).then(pl.col("store_primary_category_name")).otherwise(pl.col("store_primary_category")).alias("store_primary_category")
).drop("store_primary_category_name")
```

In [32]:
```
market_id_per_store = df.group_by("store_id").agg(
    pl.col("market_id").mode().first().alias("market_id_num")
)
market_id_per_store = market_id_per_store.fill_null(pl.col("market_id_num").mode().first())
```

In [33]:
```
df = df.join(market_id_per_store, on="store_id", how="left")
```

In [34]:
```
df = df.with_columns(
   pl.when(pl.col("market_id").is_null()).then(pl.col("market_id_num")).otherwise(pl.col("market_id")).alias("market_id")
).drop("market_id_num")
```

In [35]:
```
df.null_count()
```

Out[35]: shape: (1, 14)

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshift_partners	total_busy_partne
u32	u32	u32	u32	u32	u32	u32	u32	u32	u32	u32	u32	u
0	0	0	0	0	904	0	0	0	0	0	0	

In [36]:
```
df = df.with_columns(
    pl.col("created_at").str.to_datetime(),
    pl.col("actual_delivery_time").str.to_datetime()
)
```

Time based feature generation

In [40]:
```
df=df.with_columns(
    pl.col("created_at").dt.offset_by('-5h30m'),
    pl.col("actual_delivery_time").dt.offset_by('-5h30m')
)
```

In [41]:
```
df = df.with_columns(
    pl.col("created_at").dt.year().alias("created_at_year"),
    pl.col("created_at").dt.month().alias("created_at_month"),
    pl.col("created_at").dt.day().alias("created_at_day"),
    pl.col("created_at").dt.hour().alias("created_at_hour"),
    pl.col("created_at").dt.minute().alias("created_at_minute"),
    pl.col("created_at").dt.second().alias("created_at_second"),
    pl.col("actual_delivery_time").dt.year().alias("actual_delivery_time_year"),
    pl.col("actual_delivery_time").dt.month().alias("actual_delivery_time_month"),
    pl.col("actual_delivery_time").dt.day().alias("actual_delivery_time_day"),
    pl.col("actual_delivery_time").dt.hour().alias("actual_delivery_time_hour"),
    pl.col("actual_delivery_time").dt.minute().alias("actual_delivery_time_minute"),
    pl.col("actual_delivery_time").dt.second().alias("actual_delivery_time_second")
)
```

In [42]:
```
df
```

Out[42]: shape: (177_544, 26)

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshi
f64	datetime[μs]	datetime[μs]	str	str	f64	i64	i64	i64	f64	f64	
1.0	2015-02-06 16:54:17	2015-02-06 17:57:16	"df263d996281d984952c07998dc543...	"american"	1.0	4	3441	4	557.0	1239.0	
2.0	2015-02-10 16:19:25	2015-02-10 17:26:29	"f0ade77b43923b38237db569b016ba...	"mexican"	2.0	1	1900	1	1400.0	1400.0	
3.0	2015-01-22 15:09:28	2015-01-22 15:39:09	"f0ade77b43923b38237db569b016ba...	"other"	1.0	1	1900	1	1900.0	1900.0	
3.0	2015-02-03 15:51:45	2015-02-03 16:43:00	"f0ade77b43923b38237db569b016ba...	"other"	1.0	6	6900	5	600.0	1800.0	
3.0	2015-02-14 21:10:36	2015-02-14 21:50:26	"f0ade77b43923b38237db569b016ba...	"other"	1.0	3	3900	3	1100.0	1600.0	
...	...	...	...	...	...	...	...	...	...	...	
1.0	2015-02-16 18:49:41	2015-02-16 19:54:48	"a914ecef9c12ffdb9bede64bb703d8...	"fast"	4.0	3	1389	3	345.0	649.0	
1.0	2015-02-12 18:31:59	2015-02-12 19:28:22	"a914ecef9c12ffdb9bede64bb703d8...	"fast"	4.0	6	3010	4	405.0	825.0	
1.0	2015-01-23 23:16:08	2015-01-24 00:06:16	"a914ecef9c12ffdb9bede64bb703d8...	"fast"	4.0	5	1836	3	300.0	399.0	
1.0	2015-02-01 12:48:15	2015-02-01 13:53:22	"c81e155d85dae5430a8cee6f2242e8...	"sandwich"	1.0	1	1175	1	535.0	535.0	
1.0	2015-02-08 13:54:33	2015-02-08 14:31:41	"c81e155d85dae5430a8cee6f2242e8...	"sandwich"	1.0	4	2605	4	425.0	750.0	

In [43]:
```
cols=[
 'order_protocol',
 'total_items',
 'subtotal',
 'num_distinct_items',
 'min_item_price',
 'max_item_price',
 'total_onshift_partners',
```

```
'total_busy_partners',
'total_outstanding_orders',
'created_at_year',
'created_at_month',
'created_at_day',
'created_at_hour',
'created_at_minute',
'actual_delivery_time_year',
'actual_delivery_time_month',
'actual_delivery_time_day',
'actual_delivery_time_hour',
'actual_delivery_time_minute'
]
```

In [44]:
```
# scaler = StandardScaler()
# scaled_features = scaler.fit_transform(df[cols].to_numpy())
knn_imputer = KNNImputer(n_neighbors=3)
imputed_features = knn_imputer.fit_transform(df[cols].to_numpy())
```

In [45]:
```
# imputed_features = scaler.inverse_transform(imputed_features)
imputed_features = pl.DataFrame(imputed_features)
imputed_features.columns = cols
imputed_features
```

Out[45]: shape: (177_544, 19)

order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshift_partners	total_busy_partners	total_outstanding_orders	created_at_year	created_at_month	created_at_d
f64	f64	f64	f64	f64	f64	f64	f64	f64	f64	f64	f
1.0	4.0	3441.0	4.0	557.0	1239.0	33.0	14.0	21.0	2015.0	2.0	
2.0	1.0	1900.0	1.0	1400.0	1400.0	1.0	2.0	2.0	2015.0	2.0	1
1.0	1.0	1900.0	1.0	1900.0	1900.0	1.0	0.0	0.0	2015.0	1.0	2
1.0	6.0	6900.0	5.0	600.0	1800.0	1.0	1.0	2.0	2015.0	2.0	
1.0	3.0	3900.0	3.0	1100.0	1600.0	6.0	6.0	9.0	2015.0	2.0	1
…	…	…	…	…	…	…	…	…	…	…	…
4.0	3.0	1389.0	3.0	345.0	649.0	17.0	17.0	23.0	2015.0	2.0	1
4.0	6.0	3010.0	4.0	405.0	825.0	12.0	11.0	14.0	2015.0	2.0	1
4.0	5.0	1836.0	3.0	300.0	399.0	39.0	41.0	40.0	2015.0	1.0	2
1.0	1.0	1175.0	1.0	535.0	535.0	7.0	7.0	12.0	2015.0	2.0	
1.0	4.0	2605.0	4.0	425.0	750.0	20.0	20.0	23.0	2015.0	2.0	

In [46]:
```
df=df.with_columns(
    pl.when(pl.col("order_protocol").is_null())
    .then(imputed_features["order_protocol"])
    .otherwise(pl.col("order_protocol"))
    .cast(pl.Int8)
    .alias("order_protocol")
)
```

In [47]:
```
df.write_parquet("../data/cleaned/dataset.parquet")
```

Feature Generation

In [54]:
```
df=pl.read_parquet("../data/cleaned/dataset.parquet")
```

In [55]:
```
df["created_at"].max(), df["created_at"].min()
```

Out[55]:
```
(datetime.datetime(2015, 2, 18, 0, 30, 44),
 datetime.datetime(2015, 1, 21, 9, 52, 3))
```

In [56]:
```
df=df.with_columns(
    pl.col("created_at").dt.strftime("%b").alias("created_at_month_name"),
    pl.col("actual_delivery_time").dt.strftime("%b").alias("actual_delivery_time_month_name"),
    pl.col("created_at").dt.weekday().alias("created_at_day_of_week"),
    pl.col("actual_delivery_time").dt.weekday().alias("actual_delivery_time_day_of_week"),
    pl.col("created_at").dt.strftime("%a").alias("created_at_day_name"),
    pl.col("actual_delivery_time").dt.strftime("%a").alias("actual_delivery_time_day_name"),
    pl.col("created_at").dt.week().alias("created_at_week_number"),
    pl.col("actual_delivery_time").dt.week().alias("actual_delivery_time_week_number"),
   ((pl.col("created_at").dt.day() - 1) // 7 + 1).cast(pl.Int16).alias("created_at_week_of_month"),
    ((pl.col("actual_delivery_time").dt.day() - 1) // 7 + 1).cast(pl.Int16).alias("actual_delivery_time_week_of_month")

)
```

In [57]:
```
df=df.with_columns(
    ((pl.col("actual_delivery_time") - pl.col("created_at")).dt.total_seconds())
    .alias("delivery_time_seconds"),
    ((pl.col("actual_delivery_time") - pl.col("created_at")).dt.total_minutes())
    .alias("delivery_time_minutes"),
)
```

In [58]:
```
df.with_columns(
    pl.col("market_id").cast(pl.Int8).alias("market_id"),
    pl.col("total_items").cast(pl.Int16).alias("total_items"),
    pl.col("subtotal").cast(pl.Int16).alias("subtotal"),
    pl.col("num_distinct_items").cast(pl.Int16).alias("num_distinct_items"),
    pl.col("min_item_price").cast(pl.Int16).alias("min_item_price"),
    pl.col("max_item_price").cast(pl.Int16).alias("max_item_price"),
)
```

Out[58]: shape: (177_544, 38)

market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshi
i8	datetime[µs]	datetime[µs]	str	str	i8	i16	i16	i16	i16	i16	
1	2015-02-06 16:54:17	2015-02-06 17:57:16	"df263d996281d984952c07998dc543...	"american"	1	4	3441	4	557	1239	
2	2015-02-10 16:19:25	2015-02-10 17:26:29	"f0ade77b43923b38237db569b016ba...	"mexican"	2	1	1900	1	1400	1400	
3	2015-01-22 15:09:28	2015-01-22 15:39:09	"f0ade77b43923b38237db569b016ba...	"other"	1	1	1900	1	1900	1900	
3	2015-02-03 15:51:45	2015-02-03 16:43:00	"f0ade77b43923b38237db569b016ba...	"other"	1	6	6900	5	600	1800	
3	2015-02-14 21:10:36	2015-02-14 21:50:26	"f0ade77b43923b38237db569b016ba...	"other"	1	3	3900	3	1100	1600	
...	...	...	...	...	...	...	...	...	...	...	
1	2015-02-16 18:49:41	2015-02-16 19:54:48	"a914ecef9c12ffdb9bede64bb703d8...	"fast"	4	3	1389	3	345	649	
1	2015-02-12 18:31:59	2015-02-12 19:28:22	"a914ecef9c12ffdb9bede64bb703d8...	"fast"	4	6	3010	4	405	825	
1	2015-01-23 23:16:08	2015-01-24 00:06:16	"a914ecef9c12ffdb9bede64bb703d8...	"fast"	4	5	1836	3	300	399	
1	2015-02-01 12:48:15	2015-02-01 13:53:22	"c81e155d85dae5430a8cee6f2242e8...	"sandwich"	1	1	1175	1	535	535	
1	2015-02-08 13:54:33	2015-02-08 14:31:41	"c81e155d85dae5430a8cee6f2242e8...	"sandwich"	1	4	2605	4	425	750	

In [59]:
```
df.write_parquet("../data/cleaned/dataset.parquet")
```